

Topic 1: Descriptive Analysis of Education and Employment in German Districts

Data

The data for this project are based on the 2022 German Census (*Zensus 2022*) conducted by the Federal Statistical Office of Germany. The raw data can be downloaded from the following link:

https://www.destatis.de/static/DE/zensus/gitterdaten/Regionaltabelle_Bildung_Erwerbstaetigkeit.xlsx

The accompanying R script *clean_zensus2022.R* processes the raw data and generates two analysis files: *zensus2022_wide.csv* and *zensus2022_long.csv*. Please run the script before starting your analysis. The file *zensus2022_wide.csv* contains information on educational attainment and employment status for 400 districts and independent cities (*Kreise und kreisfreie Städte*) across Germany. The variables are:

- **ARS:** Official regional key (five-digit code);
- **Name:** Name of the district or independent city;
- **Bundesland:** Abbreviation of the federal state (e.g. NRW, BY);
- **Ostdeutschland:** Dichotomous variable; 1 = eastern federal state (BB, BE, MV, SN, ST, TH), 0 = western federal state;
- **Bev_Tausend:** Population aged 15 and over (in thousands);
- **Abitur_Quote:** Share of persons with a university entrance qualification (*Abitur* or *Fachhochschulreife*) among all persons with a school-leaving certificate (%);
- **OhneAbschl_Quote:** Share of persons without a general school-leaving certificate among the population aged 15 and over (%);
- **Abitur_GenderGap:** Difference in the *Abitur* rate between women and men (in percentage points; positive = women higher);
- **Erwerbstaetig_Quote:** Share of employed persons among the population aged 15 and over (%);
- **Erwerbslos_Quote:** Share of unemployed persons among the labour force (%; corresponding to the standard unemployment rate);
- **Nichterwerb_Quote:** Share of economically inactive persons among the population aged 15 and over (%);

- **Erwerbstaetig_GenderGap:** Difference in the employment rate between women and men (in percentage points);
- **Erwerbslos_GenderGap:** Difference in the unemployment rate between women and men (in percentage points).

The file *zensus2022_long.csv* contains the same information in long format (one row per district and variable) and is particularly suited for compact visualisations using `facet_wrap`.

Tasks

Start by reading in the data. Then proceed as follows:

1. Does the dataset require any cleaning before you can work with it? Do any variables need to be converted or otherwise prepared? *Hint:* Are the scales of the variables appropriate? Which variables are metric, which are categorical?
2. Are there outliers, and if so: how might you deal with them? Which districts or cities stand out in particular?
3. Examine the univariate distributions of all variables separately for East and West Germany. Use both histograms and boxplots. What do you notice?
4. Examine pairwise scatterplots of the metric variables. What do you observe? Use the variable `Ostdeutschland` to colour the points.
5. Investigate the pairwise correlations between the metric variables and produce a correlation plot. Assess the magnitude of the correlations and test their statistical significance. Research conventions for interpreting correlation coefficients. *Hint:* `cor.plot` from the R package `psych`.
6. Discuss: based on your results so far, can you conclude which variables causally contribute to high (or low) values of the others?
7. Now describe more systematically: identify at least three variables that have the potential to influence the unemployment rate. Consider which univariate and multivariate methods would be appropriate.
8. Compare East and West Germany systematically using descriptive statistics (e.g. median, standard deviation). What structural differences can still be observed more than three decades after reunification?
9. You have likely made several interesting observations. Make a deliberate selection of results to include in your report: is there an overarching question or topic of broad public interest that could motivate your report? Please also refer to the reporting guidelines in Moodle.

Notes

- You may use the `ggplot2` package in R to produce high-quality visualisations.

- Useful R functions: `readr::read_csv`, `str`, `summary`, `hist`, `plot`, `boxplot`, `table`, `prop.table`, `psych::describe`, `psych::cor.plot`, `tidyr::pivot_longer`, `ggplot2::facet_wrap`.
- The raw data are publicly available from the Federal Statistical Office of Germany (Zensus 2022, reference date: 15 May 2022) at <https://www.destatis.de>.

Please submit your report in electronic form together with your R code (including all necessary files) in Moodle by Wednesday, 29 April 2026, 23:59. A printed copy is not required.